pyAmpliCol at one loop

Benchmarks of the recursive one-loop engine against MadLoop, Recola and OpenLoops

report 2026-11-12 · finite part, binary64, one pinned thread, $B = 128$

Table 1: engine `524e6cfa`, timed 2026-11-11, AMD EPYC 9754, shared host · Table 2: engine `524e6cfa`

Code: pyAmpliCol 1.0.3, commit `779f4931` (TIRust `bc94b496`, NinjaRS `5fe7f392`, oneloopmaster `bd703d09`, Symbolica `dfe6bd97`). Times measured at `524e6cfa` (TIRust `fefb5f2b`, NinjaRS `71eddb15`, oneloopmaster `bd703d09`, Symbolica `dfe6bd97`); the times are the same at the shipped commit (paired same-core check on two rows, B = 128: ×0.98 and ×1.01, within the timing noise), and Table 3's loop-induced agreement check is measured there. Older measurements kept: Table 3's Born-interfered agreement check (2026-11-08); the validation of the loop-induced processes where TIRust served every point (not re-run) (2026-11-08); the double-double and arbitrary-precision checks of g g → a a g g, g g → h g g g, g g → h h g g and g g → z z z z (multi-hour, not re-run) (2026-11-08).

×1/16 faster · ×1/4 · ×1 · ×4 · ×16 slower — run time ÷ fastest provider

Gate pass ×0.25 · fail ×1.4 · open ×1.0 · <5 ms: not gated ×1.4 — time ÷ limit; limit = min(ML, 1.30 × OL, 1.30 × Rc)

Stages inputs · numerator · scatter · NinjaRS · masters · glue · missing (why) · no colour: generation, LC · † within estimate · ¶ reference's own (DD) · ‡ accepted above tolerance (notes) · ML MadLoop · Rc Recola · OL OpenLoops

Table 1: **Born-interfered one-loop processes**: the one-loop amplitude interfered with the Born. Key above. ≤ before a provider time: an upper bound, from a run outside the protocol that does more than the production work ($u\bar{u} \to ggggg$); the multiplier is then a lower bound (≥) and sets no gate limit. Notes in Reading the tables.

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	pyAmpliCol (newest measurement of each cell) · 2 → 2 to 2 → 5: engine `524e6cfa`, native build `loop-native-v2-slice-678c4749 (9de26a2e…)`, timed 2026-11-11 · AMD EPYC 9754, shared host			
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$; provider times; gate chip (time ÷ limit); stage shares	median per point; LC/FC = LC time ÷ full-colour time; no provider ratio
2 → 2	$gg \to gg$ 4 legs · QCD · 24 families (LC 12) Rc·OL ≤ 3.0e−13 · ML 1.3e−12 16 pts · median 2.4e−14 · †	**9.32 s** ×36 0.57 GiB peak · plan 0.0240 MB **Rc 258 ms** · ML 9.70 s	**1.44 ms** ×0.68 **ML 2.13 ms** · OL 7.61 ms · Rc 18.9 ms $B = 128$ no gate (2→2) numerator 43 %	**0.909 ms** LC/FC **0.63** $B = 128$ · FC 1.44 ms
2 → 3	$u\bar{u} \to d\bar{d}g$ 5 legs · QCD · 42 families (LC 27) ML·OL·Rc ≤ 3.0e−12 16 pts · median 6.9e−14	**10.4 s** ×13 0.58 GiB peak · plan 0.0300 MB **Rc 783 ms** · ML 9.87 s	**3.02 ms** ×0.84 **OL 3.60 ms** · Rc 5.86 ms · ML 5.93 ms $B = 128$ <5 ms: not gated ×0.64 numerator 34 %	**2.03 ms** LC/FC **0.67** $B = 128$ · FC 3.02 ms
	$u\bar{u} \to u\bar{u}g$ 5 legs · QCD · 68 families (LC 42) OL·ML·Rc ≤ 6.0e−12 16 pts · median 1.7e−13	**13.7 s** ×27 0.58 GiB peak · plan 0.0360 MB **Rc 503 ms** · ML 10.9 s	**2.43 ms** ×0.64 **OL 3.82 ms** · Rc 5.41 ms · ML 9.03 ms $B = 128$ <5 ms: not gated ×0.49 numerator 38 %	**1.50 ms** LC/FC **0.62** $B = 128$ · FC 2.43 ms
	$u\bar{u} \to e^-e^+g$ 5 legs · QCD · 11 families (LC 7) ML·Rc ≤ 7.5e−14 16 pts · median 1.4e−14	**17.4 s** ×21 0.61 GiB peak · plan 0.0530 MB **Rc 815 ms** · ML 14.0 s	**0.672 ms** ×1.1 **Rc 0.593 ms** · ML 0.945 ms · OL no library $B = 128$ <5 ms: not gated ×0.87 numerator 34 %	**0.499 ms** LC/FC **0.74** $B = 128$ · FC 0.672 ms
	$u\bar{u} \to ggg$ 5 legs · QCD · 90 families (LC 62) Rc·ML ≤ 9.2e−13 · OL 8.4e−11 16 pts · median 5.2e−14 · ¶	**11.8 s** ×11 0.58 GiB peak · plan 0.0490 MB **Rc 1.06 s** · ML 17.6 s	**6.73 ms** ×0.30 **OL 22.4 ms** · Rc 46.0 ms · ML 52.2 ms $B = 128$ pass ×0.23 numerator 43 %	**3.18 ms** LC/FC **0.47** $B = 128$ · FC 6.73 ms
	$gg \to ggg$ 5 legs · QCD · 154 families (LC 77) Rc 1.4e−12 · OL·ML ≤ 3.0e−6 16 pts · median 5.8e−13 · †	**12.3 s** ×1.1 0.60 GiB peak · plan 0.112 MB **Rc 11.7 s** · ML 47.0 s	**24.6 ms** ×0.14 **OL 171 ms** · ML 287 ms · Rc 784 ms $B = 128$ pass ×0.11 numerator 40 %	**11.3 ms** LC/FC **0.46** $B = 128$ · FC 24.6 ms
2 → 4	$u\bar{u} \to e^-e^+gg$ 6 legs · QCD · 69 families (LC 33) Rc 3.7e−13 · ML 1.9e−10 16 pts · median 7.2e−14 · †	**19.6 s** ×13 0.62 GiB peak · plan 0.0680 MB **Rc 1.48 s** · ML 24.8 s	**6.14 ms** ×0.33 **Rc 18.9 ms** · ML 61.3 ms · OL no library $B = 128$ pass ×0.25 numerator 48 %	**2.72 ms** LC/FC **0.44** $B = 128$ · FC 6.14 ms
	$gg \to t\bar{t}gg$ 6 legs · QCD · 1 086 families (LC 375) Rc 1.3e−11 · OL·ML ≤ 6.6e−9 16 pts · median 1.2e−11 · †	**19.3 s** ×0.30 0.64 GiB peak · plan 0.320 MB **Rc 63.9 s** · ML 218 s	**76.1 ms** ×0.15 **OL 501 ms** · Rc 2.19 s · ML 5.65 s $B = 128$ pass ×0.12 numerator 37 %	**28.1 ms** LC/FC **0.37** $B = 128$ · FC 76.1 ms
	$u\bar{u} \to gggg$ 6 legs · QCD · 628 families (LC 375) OL·Rc ≤ 2.8e−10 16 pts · median 1.7e−12 · †	**23.4 s** ×0.65 0.66 GiB peak · plan 0.382 MB **Rc 35.8 s**	**58.7 ms** ×0.15 **OL 398 ms** · Rc 1.23 s · ML no production timing $B = 128$ pass ×0.11 numerator 48 %	**26.7 ms** LC/FC **0.46** $B = 128$ · FC 58.7 ms
	$gg \to gggg$ 6 legs · QCD · 1 100 families (LC 550) Rc·OL·ML ≤ 1.7e−7 8 pts · median 2.3e−11 · †	**20.2 s** ×0.0095 0.90 GiB peak · plan 1.23 MB **Rc 35.4 min** · ML 144.6 min	**463 ms** ×0.097 **OL 4.80 s** · Rc 17.6 s[p] · ML 56.9 s $B = 128$ pass ×0.074 inputs 42 %	**175 ms** LC/FC **0.38** $B = 128$ · FC 463 ms

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
pyAmpliCol (newest measurement of each cell) · $2 \to 2$ to $2 \to 5$: engine 524e6cfa, native build loop-native-v2-slice-678c4749 (9de26a2e…), timed 2026-11-11 · AMD EPYC 9754, shared host				
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$; provider times; gate chip (time ÷ limit); stage shares	median per point; LC/FC = LC time ÷ full-colour time; no provider ratio
$2 \to 5$	$u\bar{u} \to e^- e^+ d\bar{d} g$ 7 legs · QCD · 242 families (LC 114) OL·ML·Rc ≤ 4.5e−11 16 pts · median 8.9e−13	**14.8 s** ×2.1 0.63 GiB peak · plan 0.0960 MB **Rc 6.95 s** · ML 58.6 s	**14.3 ms** ×0.37 **Rc 38.2 ms** · OL 76.4 ms · ML 167 ms $B = 128$ pass ×0.29 numerator 43 %	**6.51 ms** LC/FC **0.46** $B = 128$ · FC 14.3 ms
	$u\bar{u} \to e^- e^+ ggg$ 7 legs · QCD · 463 families (LC 164) Rc·OL·ML ≤ 6.1e−11 16 pts · median 2.3e−12 · †	**14.8 s** ×1.1 0.64 GiB peak · plan 0.179 MB **Rc 13.9 s** · ML 146 s	**43.2 ms** ×0.15 **Rc 286 ms** · OL 363 ms · ML 1.48 s $B = 128$ pass ×0.12 numerator 58 %	**9.38 ms** LC/FC **0.22** $B = 128$ · FC 43.2 ms
	$gg \to t\bar{t} e^- e^+ g$ 7 legs · QCD · 579 families (LC 164) Rc 2.6e−10 · OL·ML ≤ 1.7e−8 16 pts · median 1.7e−11 · †	**20.7 s** ×0.64 0.65 GiB peak · plan 0.176 MB **Rc 32.3 s** · ML 165 s	**75.4 ms** ×0.10 **OL 726 ms** · Rc 854 ms · ML 4.02 s $B = 128$ pass ×0.080 numerator 48 %	**15.5 ms** LC/FC **0.21** $B = 128$ · FC 75.4 ms
	$gg \to t\bar{t} e^- e^+ g$ 7 legs · EW · 1 599 families OL 2.7e−10 · ML 2.8e−9 16 pts · median 7.6e−10	**453 s** ×13 9.40 GiB peak · plan 0.445 MB **Rc 34.2 s** · ML 11.9 min	**181 ms** ×0.12 **OL 1.52 s** · Rc 2.09 s · ML 11.3 s $B = 128$ pass ×0.091 numerator 39 % model: unpatched G_mu UFO	***not supported*** refused by design: no leading-colour expansion for electroweak corrections
	$u\bar{u} \to ggggg$ 7 legs · QCD · 5 258 families (LC 2 712) ML 2.1e−12 · Rc 2.9e−10 3 pts · median 7.0e−11 · ¶	**136 s** ×0.013 2.76 GiB peak · plan 14.7 MB **Rc 170.3 min** · ML 12.2 h	**1.81 s** ≥×0.0099 **Rc ≤182 s** · ML 569 s^p · OL no library $B = 128$ pass ×0.0032 numerator 52 %	**596 ms** LC/FC **0.33** $B = 128$ · FC 1.81 s
	$gg \to ggggg$ 7 legs · QCD · 4 707 families (LC 4 707) NJet DD 1.5e−7 10 pts · median 3.4e−9 · † · top-free (particle_veto = [t]), Nf = 5 ‡ pts 4, 8 above 2.0e−8: accepted	**173 s** 4.39 GiB peak · plan 20.2 MB ML diagrams 17.4 h, stopped in export · Rc unfinished	**5.11 s** $B = 128$ no gate (no matched provider) inputs 46 %	**4.01 s** LC/FC **0.78** $B = 128$ · FC 5.11 s
$2 \to 6$: pyAmpliCol only (no provider exists): no ratio, gate or colour · engine 2f3f76d9, native build loop-native-mf-2to6-2f3f76d9-lto (2bae17f9…), older than the engine above, not re-run · shared host, indicative · internal checks only				
$2 \to 6$	$gg \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 1.0e−11 · E ≤ 1.4e−3 (2 pts above 10^{-5}) · binary64 = DD to 7.4e−5 · rescue 11/16 · NGluon DD to 6.3e−13 · **FC** ε^{-2} ≤ 8.4e−11 · E ≤ 5.1e−3 (2 pts above 10^{-5}) · Catani ε^{-1} ≤ 7.1e−5 · binary64 = DD to 1.2e−4 · rescue 11/16 Ward FC not run, Ward LC not asserted	**55.3 min** FC 74.0 GiB peak · plan 486 MB · store 25.8 min artifact 0.53 GB · load 4.21 s (1.06 GiB) **LC** 46.1 min · 63.9 GiB · artifact 0.364 GB	**201 s** 1 block of 16 $B = 16$ no provider: no gate inputs 55 % peak 40.1 GiB	**98.0 s** 1 block of 128 $B = 128$ numerator 54 % peak 27.2 GiB
	$u\bar{u} \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 4.6e−12 · E ≤ 6.2e−7 · binary64 = DD to 8.4e−11 · rescue 1/16 · Ward ≤ 6.5e−12 · **FC** ε^{-2} ≤ 1.1e−11 · E ≤ 5.9e−6 · Catani ε^{-1} ≤ 3.3e−8 · binary64 = DD to 1.7e−10 · rescue 2/16 · Ward ≤ 2.1e−11	**73.6 min** FC 92.9 GiB peak · plan 814 MB · store 449 s artifact 0.143 GB · load 3.85 s (1.66 GiB) **LC** 20.5 min · 47.0 GiB · artifact 0.047 GB	**62.1 s** 1 block of 128 $B = 128$ no provider: no gate numerator 77 % peak 31.7 GiB	**10.9 s** 1 block of 128 $B = 128$ numerator 60 % peak 12.6 GiB
	$u\bar{u} \to u\bar{u}gggg$ 8 legs · QCD · 40 454 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 6.7e−14 · E ≤ 7.5e−7 · binary64 = DD to 1.3e−11 · rescue 2/16 · Ward ≤ 4.3e−12 · **FC** ε^{-2} ≤ 2.0e−13 · E ≤ 6.5e−6 · Catani ε^{-1} ≤ 6.6e−11 · binary64 = DD to 1.3e−11 · rescue 4/16 · Ward ≤ 8.5e−12 (DD 1.1e−28)	**29.8 min** FC 42.5 GiB peak · plan 315 MB · store 46.4 s artifact 0.038 GB · load 1.51 s (0.68 GiB) **LC** 506 s · 18.2 GiB · artifact 0.009 GB	**19.2 s** 1 block of 128 $B = 128$ no provider: no gate numerator 75 % peak 36.0 GiB	**2.67 s** 1 block of 128 $B = 128$ numerator 55 % peak 7.38 GiB

Table 2: **Loop-induced processes**: $|A_{\text{loop}}|^2$, no Born; read as Table 1. Timed with the unchecked reduction setting[1]. Notes in Reading the tables.

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
legs · corrections; largest deviation per reference (AN: analytic result) · verdict · points	public `Generator`, warm cache (cold: empty cache); peak memory; provider generation	median per point at $B = 128$ over fresh runtimes (× spread); paired provider times; gate chip on TIRust; backend (`reduction=` value); stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (separate instrumented run, not the timed call)

pyAmpliCol: engine 524e6cfa, native build `loop-native-v2-slice-678c4749 (9de26a2e…)` on the host of Table 1 · cell: TIRust unchecked (= `auto` unchecked, paired on $gg \to hgg$), NinjaRS unchecked beneath · gate on TIRust · measured 2026-11-11 · host state: up to 41 rows timed in parallel, 1-min load 203–630, MemAvailable ≥ 247.9 GiB, CPU share of the timed calls ≥ 0.838 (median per row), 17 rows timed again on another core (disturbed first sample or regenerated stream)
Protocol: `reduction_unchecked = true (default false)`, explicit `tirust` and `ninjars`; `poles="none"`, `on_failure="return"`, no stability check, no rescue; the 128 distinct stream points per call, one thread on one core (`taskset`, `RAYON_NUM_THREADS=1`); 3 fresh runtimes per arm in rotated order, median of their medians over rounds (spread ×: largest ÷ smallest); providers in the same unit and on the same core, alternating with pyAmpliCol; engine 524e6cfa replication: `docs/performance_reports/loop/README.md (unchecked Table 2)`

$2 \to 1 \cdot 1 \to 2$

$gg \to h$ 3 legs · QCD AN 2.6e−13 · OL 6.2e−13 · ML 9.2e−13 · Rc 9.3e−13 · pass · 6 pts	**12.4 s** cold 17.1 s · 0.63 GiB peak warm 0.61 GiB peak · plan 0.0100 MB	**0.345 ms** ×4.9 **Rc 0.0705 ms** · ML 0.0726 ms · OL 0.177 ms $B = 128$ · 3 runtimes ×1.0 no gate (2→1) **TIRust** (`tirust`) glue 85 % vs 256-bit truth (128 pts): max 1.2e−13 **NinjaRS per amplitude** (`ninjars`) 0.363 ms ×5.2 glue 80 % vs 256-bit truth (128 pts): max 1.2e−12
$h \to gg$ 3 legs · QCD AN 6.2e−14 · Rc 6.3e−13 · ML 6.3e−13 · pass · 3 pts	**9.62 s** cold 16.4 s · 0.63 GiB peak warm 0.62 GiB peak · plan 0.0100 MB	**0.368 ms** ×5.0 **Rc 0.0731 ms** · ML 0.0805 ms · OL unavailable: a decay: OpenLoops' loop-squared libraries cover 2 → n scattering only $B = 128$ · 3 runtimes ×1.1 no gate (1→2) **TIRust** (`tirust`) glue 84 % vs 256-bit truth (128 pts): max 1.0e−13 **NinjaRS per amplitude** (`ninjars`) 0.386 ms ×5.3 glue 80 % vs 256-bit truth (128 pts): max 1.1e−12
$h \to aa$ 3 legs · QED AN 2.9e−14 · ML 1.9e−13 · Rc 2.0e−13 · pass · 3 pts	**114 s** cold 190 s · 3.32 GiB peak warm 2.71 GiB peak · plan 0.00400 MB	**0.668 ms** ×6.8 **Rc 0.0975 ms** · ML 0.157 ms · OL unavailable: a decay: OpenLoops' loop-squared libraries cover 2 → n scattering only $B = 128$ · 3 runtimes ×1.0 no gate (1→2) **TIRust** (`tirust`) inputs 70 % vs 256-bit truth (128 pts): max 2.4e−14 **NinjaRS per amplitude** (`ninjars`) 0.717 ms ×7.4 inputs 74 % vs 256-bit truth (128 pts): max 4.2e−13
$h \to za$ 3 legs · QED ML 1.6e−14 · AN 1.7e−14 · Rc 7.9e−14 · pass · 3 pts	**154 s** cold 231 s · 3.50 GiB peak warm 2.98 GiB peak · plan 0.00400 MB	**1.000 ms** ×4.5 **Rc 0.220 ms** · ML 0.274 ms · OL unavailable: a decay: OpenLoops' loop-squared libraries cover 2 → n scattering only $B = 128$ · 3 runtimes ×1.0 no gate (1→2) **TIRust** (`tirust`) inputs 68 % vs 256-bit truth (128 pts): max 1.2e−14 **NinjaRS per amplitude** (`ninjars`) 1.06 ms ×4.8 inputs 72 % vs 256-bit truth (128 pts): max 1.6e−12
$aa \to h$ 3 legs · QED AN 5.2e−14 · ML 2.2e−13 · Rc 2.2e−13 · pass · 3 pts	**159 s** cold 232 s · 3.32 GiB peak warm 2.73 GiB peak · plan 0.00400 MB	**0.708 ms** ×6.2 **Rc 0.114 ms** · ML 0.157 ms · OL unavailable: no OpenLoops loop-squared library serves this process $B = 128$ · 3 runtimes ×1.0 no gate (2→1) **TIRust** (`tirust`) inputs 73 % vs 256-bit truth (128 pts): max 1.4e−14 **NinjaRS per amplitude** (`ninjars`) 0.742 ms ×6.5 inputs 74 % vs 256-bit truth (128 pts): max 1.6e−13

$2 \to 2$

$gg \to aa$ 4 legs · QCD ML 3.4e−14 · Rc 4.5e−14 · AN 4.6e−14 · OL 4.7e−14 · pass · 6 pts	**19.2 s** cold 35.8 s · 0.70 GiB peak warm 0.69 GiB peak · plan 0.0210 MB	**1.13 ms** ×0.59 **ML 1.90 ms** · OL 2.65 ms · Rc 3.99 ms $B = 128$ · 3 runtimes ×1.4 no gate (2→2) **TIRust** (`tirust`) numerator 47 % vs 256-bit truth (128 pts): max 6.6e−13 **NinjaRS per amplitude** (`ninjars`) 3.54 ms ×1.9 NinjaRS 54 % vs 256-bit truth (128 pts): max 2.4e−13
$gg \to zz$ 4 legs · QCD Rc 4.6e−12 · OL 5.9e−12 · ML 6.6e−12 · pass · 3 pts	**19.1 s** cold 29.6 s · 0.72 GiB peak warm 0.70 GiB peak · plan 0.0290 MB	**2.36 ms** ×0.73 **ML 3.25 ms** · OL 4.12 ms · Rc 5.46 ms $B = 128$ · 3 runtimes ×1.4 no gate (2→2) **TIRust** (`tirust`) numerator 52 % vs 256-bit truth (128 pts): max 5.5e−12 **NinjaRS per amplitude** (`ninjars`) 6.10 ms ×1.9 NinjaRS 61 % vs 256-bit truth (128 pts): max 8.8e−11

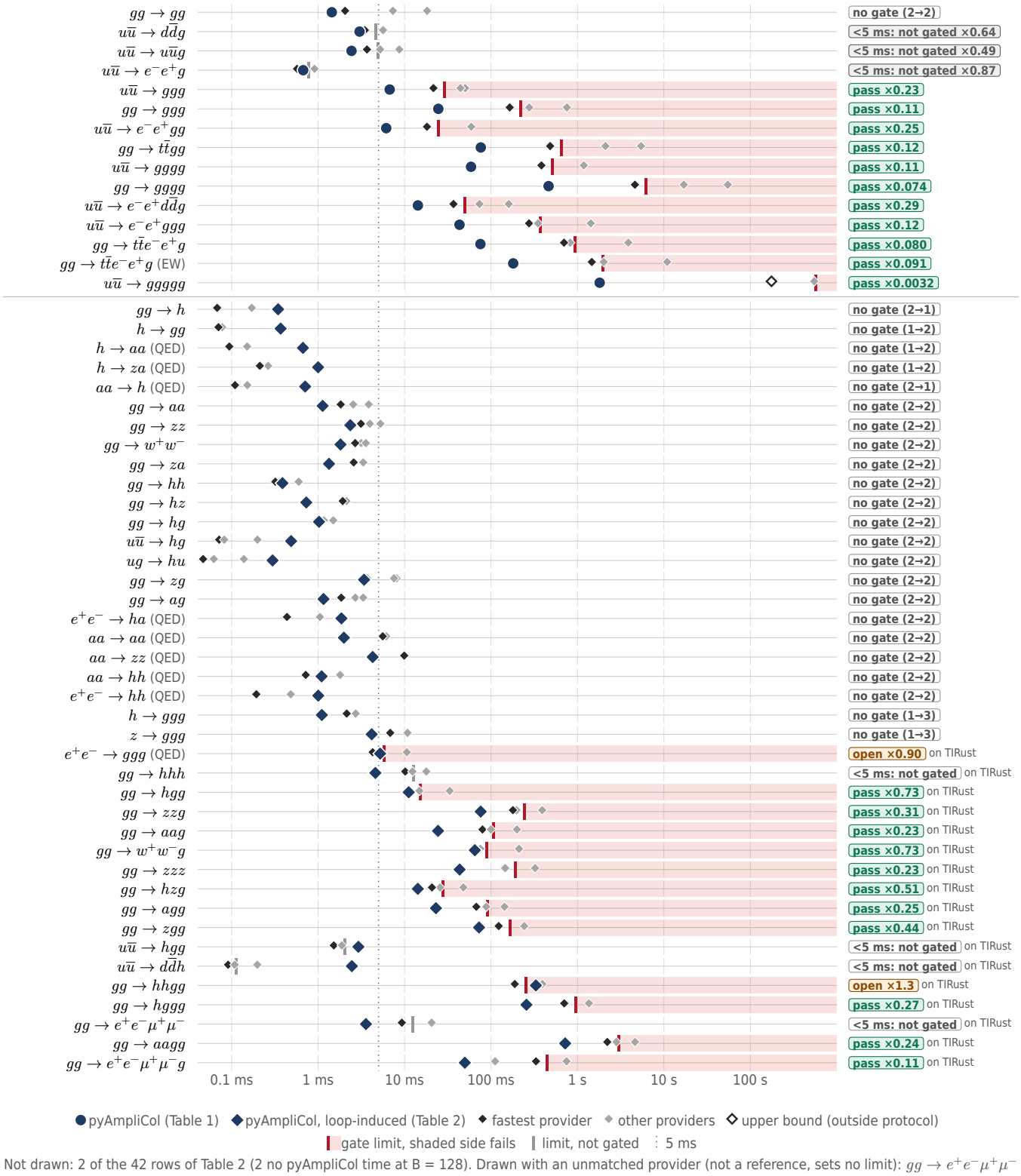
[1]Off by default. A default call (`reduction="auto"`, backend checks on) runs TIRust at the target accuracy 10^{-5} with a per-point NinjaRS fallback. With `on_failure="best-effort"` (the default), a point neither backend accepts is returned with status `best-effort`, its unchecked value and an uncertainty where the unchecked re-evaluation gives a value; otherwise it stays a failed point without a value. These times are not those of a default call.

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
legs · corrections; largest deviation per reference (AN: analytic result) · verdict · points	public `Generator`, warm cache (cold: empty cache); peak memory; provider generation	median per point at $B = 128$ over fresh runtimes (× spread); paired provider times; gate chip on TIRust; backend (`reduction=` value); stage bars: inputs, numerator, scatter, NinjaRS, TIR moments, contraction, masters, glue (separate instrumented run, not the timed call)
$2 \to 2$		
$gg \to w^+w^-$ 4 legs · QCD OL 1.7e−13 · Rc 5.7e−13 · ML 7.8e−13 · pass · 3 pts	**15.3 s** cold 19.5 s · 0.79 GiB peak warm 0.79 GiB peak · plan 0.0420 MB	**1.82 ms** **×0.65** **OL 2.79 ms** · ML 3.27 ms · Rc 3.70 ms $B = 128$ · 3 runtimes ×1.8 no gate (2→2) **TIRust** (`tirust`) numerator 46 % vs 256-bit truth (128 pts): max 1.2e−12 **NinjaRS per amplitude** (`ninjars`) 8.06 ms ×2.9 NinjaRS 66 % vs 256-bit truth (128 pts): max 5.2e−12
$gg \to za$ 4 legs · QCD Rc 1.7e−14 · OL 1.9e−14 · ML 4.2e−14 · pass · 3 pts	**10.9 s** cold 16.9 s · 0.71 GiB peak warm 0.70 GiB peak · plan 0.0360 MB	**1.34 ms** **×0.50** **OL 2.68 ms** · ML 3.38 ms · Rc 3.45 ms $B = 128$ · 3 runtimes ×1.1 no gate (2→2) **TIRust** (`tirust`) numerator 49 % vs 256-bit truth (128 pts): max 6.8e−13 **NinjaRS per amplitude** (`ninjars`) 3.82 ms ×1.4 NinjaRS 60 % vs 256-bit truth (128 pts): max 6.4e−12
$gg \to hh$ 4 legs · QCD OL 3.2e−14 · Rc 9.3e−14 · ML 1.6e−13 · pass · 3 pts	**11.7 s** **×119** cold 27.0 s · 0.64 GiB peak warm 0.63 GiB peak · plan 0.0100 MB Rc 98.4 ms	**0.387 ms** **×1.2** **ML 0.332 ms** · Rc 0.373 ms · OL 0.619 ms $B = 128$ · 3 runtimes ×1.7 no gate (2→2) **TIRust** (`tirust`) glue 48 % vs 256-bit truth (128 pts): max 3.4e−9 **NinjaRS per amplitude** (`ninjars`) 0.564 ms ×1.7 NinjaRS 41 % vs 256-bit truth (128 pts): max 8.8e−11
$gg \to hz$ 4 legs · QCD ML 5.5e−14 · Rc 6.0e−14 · OL 2.0e−13 · pass · 3 pts	**30.0 s** cold 36.6 s · 0.71 GiB peak warm 0.71 GiB peak · plan 0.0280 MB	**0.729 ms** **×0.37** **Rc 2.00 ms** · ML 2.19 ms · OL 2.21 ms $B = 128$ · 3 runtimes ×1.4 no gate (2→2) **TIRust** (`tirust`) glue 37 % vs 256-bit truth (128 pts): max 1.4e−12 **NinjaRS per amplitude** (`ninjars`) 1.26 ms ×0.63 NinjaRS 35 % vs 256-bit truth (128 pts): max 1.4e−12
$gg \to hg$ 4 legs · QCD OL 5.3e−15 · ML 7.1e−15 · Rc 1.3e−14 · pass · 3 pts	**12.7 s** cold 27.3 s · 0.65 GiB peak warm 0.63 GiB peak · plan 0.0110 MB	**1.02 ms** **×1.0** **OL 1.01 ms** · Rc 1.21 ms · ML 1.55 ms $B = 128$ · 3 runtimes ×3.1 no gate (2→2) **TIRust** (`tirust`) glue 44 % vs 256-bit truth (128 pts): max 2.4e−14 **NinjaRS per amplitude** (`ninjars`) 1.66 ms ×1.6 NinjaRS 44 % vs 256-bit truth (128 pts): max 7.3e−13
$u\bar{u} \to hg$ 4 legs · QCD Rc 2.0e−15 · ML 3.9e−15 · OL 5.1e−15 · pass · 3 pts	**12.2 s** cold 28.8 s · 0.64 GiB peak warm 0.63 GiB peak · plan 0.0100 MB	**0.487 ms** **×6.5** **ML 0.0744 ms** · Rc 0.0852 ms · OL 0.206 ms $B = 128$ · 3 runtimes ×1.0 no gate (2→2) **TIRust** (`tirust`) glue 86 % vs 256-bit truth (128 pts): max 1.1e−14 **NinjaRS per amplitude** (`ninjars`) 0.532 ms ×7.2 glue 81 % vs 256-bit truth (128 pts): max 1.5e−13
$ug \to hu$ 4 legs · QCD OL 5.0e−15 · ML 5.3e−15 · Rc 9.4e−15 · pass · 3 pts	**11.3 s** cold 17.4 s · 0.64 GiB peak warm 0.63 GiB peak · plan 0.0100 MB	**0.297 ms** **×6.1** **ML 0.0485 ms** · Rc 0.0645 ms · OL 0.144 ms $B = 128$ · 3 runtimes ×1.1 no gate (2→2) **TIRust** (`tirust`) glue 84 % vs 256-bit truth (128 pts): max 1.2e−14 **NinjaRS per amplitude** (`ninjars`) 0.315 ms ×6.5 glue 84 % vs 256-bit truth (128 pts): max 1.1e−13
$gg \to zg$ 4 legs · QCD Rc 4.3e−14 · OL 1.1e−13 · ML 1.9e−13 · pass · 3 pts	**16.1 s** cold 34.4 s · 0.71 GiB peak warm 0.71 GiB peak · plan 0.0290 MB	**3.40 ms** **×0.90** **OL 3.79 ms** · Rc 7.83 ms · ML 8.38 ms $B = 128$ · 3 runtimes ×1.3 no gate (2→2) **TIRust** (`tirust`) glue 42 % vs 256-bit truth (128 pts): max 9.4e−13 **NinjaRS per amplitude** (`ninjars`) 12.9 ms ×3.4 NinjaRS 66 % vs 256-bit truth (128 pts): max 9.7e−12
$gg \to ag$ 4 legs · QCD OL 1.6e−13 · ML 1.9e−13 · Rc 1.9e−13 · pass · 3 pts	**13.5 s** cold 19.7 s · 0.71 GiB peak warm 0.70 GiB peak · plan 0.0230 MB	**1.15 ms** **×0.60** **ML 1.92 ms** · OL 2.80 ms · Rc 3.45 ms $B = 128$ · 3 runtimes ×2.2 no gate (2→2) **TIRust** (`tirust`) numerator 47 % vs 256-bit truth (128 pts): max 2.3e−13 **NinjaRS per amplitude** (`ninjars`) 3.09 ms ×1.6 NinjaRS 65 % vs 256-bit truth (128 pts): max 1.7e−12
$e^+e^- \to ha$ 4 legs · QED Rc 1.2e−14 · ML 1.3e−14 · pass · 3 pts	**363 s** cold 463 s · 5.01 GiB peak warm 4.40 GiB peak · plan 0.00500 MB	**1.86 ms** **×4.1** **Rc 0.453 ms** · ML 1.09 ms · OL unavailable: no OpenLoops loop-squared library serves this process $B = 128$ · 3 runtimes ×1.0 no gate (2→2) **TIRust** (`tirust`) inputs 73 % vs 256-bit truth (128 pts): max 3.8e−14 **NinjaRS per amplitude** (`ninjars`) 1.98 ms ×4.4 inputs 75 % vs 256-bit truth (128 pts): max 1.9e−13

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
legs · corrections; largest deviation per reference (AN: analytic result) · verdict · points	public `Generator`, warm cache (cold: empty cache); peak memory; provider generation	median per point at $B = 128$ over fresh runtimes (× spread); paired provider times; gate chip on TIRust; backend (`reduction=` value); stage bars: inputs, numerator, scatter, NinjaRS, TIR moments, contraction, masters, glue (separate instrumented run, not the timed call)
$2 \to 2$		
$aa \to aa$ 4 legs · QED Rc 1.5e−14 · ML 1.5e−14 · AN 2.0e−14 · pass · 3 pts	**255 s** cold 309 s · 4.95 GiB peak warm 4.53 GiB peak · plan 0.00500 MB	**1.99 ms** **×0.34** **Rc 5.81 ms** · ML 6.37 ms · OL unavailable: no OpenLoops loop-squared library serves this process $B = 128$ · 3 runtimes ×1.8 no gate (2→2) **TIRust** (`tirust`) inputs 63 % vs 256-bit truth (128 pts): max 1.8e−14 **NinjaRS per amplitude** (`ninjars`) 6.78 ms ×1.2 inputs 62 % vs 256-bit truth (128 pts): max 4.9e−14
$aa \to zz$ 4 legs · QED ML 2.5e−14 · Rc 2.6e−14 · pass · 3 pts	**210 s** cold 290 s · 5.21 GiB peak warm 4.84 GiB peak · plan 0.00500 MB	**4.28 ms** **×0.42** **Rc 10.3 ms** · OL unavailable: no OpenLoops loop-squared library serves this process · ML not timed: no production timing yet $B = 128$ · 3 runtimes ×1.4 no gate (2→2) **TIRust** (`tirust`) inputs 58 % vs 256-bit truth (128 pts): max 1.5e−13 **NinjaRS per amplitude** (`ninjars`) 11.8 ms ×1.1 inputs 57 % vs 256-bit truth (128 pts): max 7.4e−13
$aa \to hh$ 4 legs · QED Rc 6.1e−13 · ML 6.1e−13 · pass · 3 pts	**132 s** cold 200 s · 3.69 GiB peak warm 3.00 GiB peak · plan 0.00400 MB	**1.10 ms** **×1.5** **Rc 0.746 ms** · ML 1.87 ms · OL unavailable: no OpenLoops loop-squared library serves this process $B = 128$ · 3 runtimes ×1.0 no gate (2→2) **TIRust** (`tirust`) inputs 74 % vs 256-bit truth (128 pts): max 8.8e−11 **NinjaRS per amplitude** (`ninjars`) 1.55 ms ×2.1 inputs 74 % vs 256-bit truth (128 pts): max 9.1e−11
$e^+e^- \to hh$ 4 legs · QED Rc 1.3e−10 · ML 1.3e−10 · pass · 3 pts	**229 s** cold 321 s · 4.73 GiB peak warm 4.32 GiB peak · plan 0.00400 MB	**1.00 ms** **×5.0** **Rc 0.200 ms** · ML 0.501 ms · OL unavailable: no OpenLoops loop-squared library serves this process $B = 128$ · 3 runtimes ×1.0 no gate (2→2) **TIRust** (`tirust`) inputs 81 % vs 256-bit truth (128 pts): max 1.3e−10 **NinjaRS per amplitude** (`ninjars`) 1.02 ms ×5.1 inputs 80 % vs 256-bit truth (128 pts): max 1.3e−10
$1 \to 3$		
$h \to ggg$ 4 legs · QCD Rc 7.8e−13 · ML 7.1e−12 · partial · 3 pts	**9.11 s** cold 21.3 s · 0.64 GiB peak warm 0.63 GiB peak · plan 0.0110 MB	**1.10 ms** **×0.50** **Rc 2.22 ms** · ML 2.82 ms · OL unavailable: a decay: OpenLoops' loop-squared libraries cover 2 → n scattering only $B = 128$ · 3 runtimes ×4.1 no gate (1→3) **TIRust** (`tirust`) glue 40 % vs 256-bit truth (128 pts): max 6.7e−13 **NinjaRS per amplitude** (`ninjars`) 1.33 ms ×0.60 NinjaRS 44 % vs 256-bit truth (128 pts): max 1.5e−12
$z \to ggg$ 4 legs · QCD ML 7.8e−12 · Rc 3.2e−11 · partial · 3 pts	**18.9 s** cold 35.7 s · 0.71 GiB peak warm 0.71 GiB peak · plan 0.0290 MB	**4.17 ms** **×0.59** **Rc 7.09 ms** · ML 11.2 ms · OL unavailable: a decay: OpenLoops' loop-squared libraries cover 2 → n scattering only $B = 128$ · 3 runtimes ×1.0 no gate (1→3) **TIRust** (`tirust`) numerator 47 % vs 256-bit truth (128 pts): max 5.5e−11 **NinjaRS per amplitude** (`ninjars`) 11.1 ms ×1.6 NinjaRS 64 % vs 256-bit truth (128 pts): max 3.8e−10
$2 \to 3$		
$e^+e^- \to ggg$ 5 legs · QED ML 1.2e−10 · Rc 1.4e−8 · partial · 3 pts	**94.3 s** **×517** cold 125 s · 2.81 GiB peak warm 2.66 GiB peak · plan 0.0380 MB Rc 182 ms	**5.21 ms** **×1.2** **Rc 4.44 ms** · ML 11.0 ms · OL unavailable: no OpenLoops loop-squared library serves this process $B = 128$ · 3 runtimes ×1.4 open ×0.90 on TIRust **TIRust** (`tirust`) glue 41 % vs 256-bit truth (128 pts): max 3.5e−7 **NinjaRS per amplitude** (`ninjars`) 9.98 ms ×2.2 NinjaRS 59 % vs 256-bit truth (128 pts): max 3.2e−10
$gg \to hhh$ 5 legs · QCD ML 2.8e−13 · OL 3.2e−13 · Rc 8.1e−12 · pass · 3 pts	**10.5 s** **×92** cold 16.7 s · 0.65 GiB peak warm 0.63 GiB peak · plan 0.0120 MB Rc 114 ms	**4.60 ms** **×0.43** **Rc 10.6 ms** · ML 12.8 ms · OL 18.5 ms $B = 128$ · 3 runtimes ×1.0 <5 ms: not gated on TIRust **TIRust** (`tirust`) TIR moments 65 % vs 256-bit truth (128 pts): max 7.6e−7 **NinjaRS per amplitude** (`ninjars`) 5.36 ms ×0.51 NinjaRS 57 % **accuracy: 2/128 pts above 1.0e−5, worst 4.5e64, points the checked default returns as best-effort**
$gg \to hgg$ 5 legs · QCD OL 2.7e−10 · ML 2.7e−10 · Rc 2.8e−10 · partial · 3 pts	**13.2 s** **×71** cold 19.1 s · 0.75 GiB peak warm 0.74 GiB peak · plan 0.0190 MB Rc 187 ms	**11.2 ms** **×0.95** **OL 11.7 ms** · Rc 15.4 ms · ML 34.5 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.73 on TIRust **TIRust** (`tirust`) numerator 43 % vs 256-bit truth (128 pts): max 9.2e−10 **NinjaRS per amplitude** (`ninjars`) 22.5 ms ×1.9 NinjaRS 63 % vs 256-bit truth (128 pts): max 8.0e−10 `auto` with the setting: values bitwise TIRust's, 0 per-point fallbacks, time ×1.00

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
legs · corrections; largest deviation per reference (AN: analytic result) · verdict · points	public `Generator`, warm cache (cold: empty cache); peak memory; provider generation	median per point at $B = 128$ over fresh runtimes (× spread); paired provider times; gate chip on TIRust; backend (`reduction=` value); stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (separate instrumented run, not the timed call)
$2 \to 3$		
$gg \to zzg$ 5 legs · QCD ML 2.0e−8 · Rc 2.0e−8 · OL 8.1e−8 · partial · 3 pts	**33.5 s** ×31 cold 51.4 s · 1.52 GiB peak warm 1.52 GiB peak · plan 0.0380 MB Rc 1.07 s	**76.0 ms** ×0.41 **Rc 186 ms** · OL 205 ms · ML 406 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.31 on TIRust **TIRust** (`tirust`) numerator 45 % **accuracy: 18/128 pts above 1.0e−5, worst 1.2e−2, all served by TIRust's Passarino-Veltman last resort** **NinjaRS per amplitude** (`ninjars`) 460 ms ×2.5 NinjaRS 82 % vs 256-bit truth (128 pts): max 5.0e−8
$gg \to aag$ 5 legs · QCD ML 2.6e−11 · Rc 2.7e−11 · OL 4.2e−11 · partial · 3 pts	**36.2 s** ×115 cold 42.0 s · 0.96 GiB peak warm 0.95 GiB peak · plan 0.0300 MB Rc 314 ms	**24.4 ms** ×0.29 **OL 82.7 ms** · Rc 104 ms · ML 207 ms $B = 128$ · 3 runtimes ×1.3 pass ×0.23 on TIRust **TIRust** (`tirust`) numerator 58 % vs 256-bit truth (128 pts): max 7.0e−6 **NinjaRS per amplitude** (`ninjars`) 170 ms ×2.1 NinjaRS 79 % **accuracy: 1/128 pts above 1.0e−5, worst 1.0e−3, also in the checked mode**
$gg \to w^+w^-g$ 5 legs · QCD ML 8.4e−10 · Rc 3.9e−9 · OL 4.3e−8 · partial · 3 pts	**45.3 s** ×30 cold 52.2 s · 2.21 GiB peak warm 2.21 GiB peak · plan 0.0580 MB Rc 1.51 s	**65.0 ms** ×0.94 **OL 68.8 ms** · Rc 78.2 ms · ML 219 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.73 on TIRust **TIRust** (`tirust`) TIR moments 43 % **accuracy: 24/128 pts above 1.0e−5, worst 1.4e−2, all served by TIRust's Passarino-Veltman last resort** **NinjaRS per amplitude** (`ninjars`) 320 ms ×4.7 NinjaRS 81 % vs 256-bit truth (128 pts): max 6.2e−8
$gg \to zzz$ 5 legs · QCD ML 9.4e−12 · OL 2.8e−11 · Rc 2.6e−10 · pass (lax) · 3 pts	**23.3 s** ×25 cold 29.8 s · 1.05 GiB peak warm 1.04 GiB peak · plan 0.0330 MB Rc 937 ms	**43.3 ms** ×0.29 **OL 147 ms** · Rc 151 ms · ML 336 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.23 on TIRust **TIRust** (`tirust`) numerator 68 % vs 256-bit truth (128 pts): max 2.3e−8 **NinjaRS per amplitude** (`ninjars`) 20/128 failed closed, status Unstable NinjaRS 77 % **accuracy: 20 pts fail (Unstable); the call is timed whole**
$gg \to hzg$ 5 legs · QCD ML 1.9e−13 · Rc 2.1e−13 · OL 2.5e−13 · pass · 3 pts	**13.1 s** ×53 cold 29.8 s · 0.76 GiB peak warm 0.75 GiB peak · plan 0.0330 MB Rc 246 ms	**14.2 ms** ×0.66 **Rc 21.5 ms** · OL 26.7 ms · ML 49.7 ms $B = 128$ · 3 runtimes ×1.2 pass ×0.51 on TIRust **TIRust** (`tirust`) numerator 54 % vs 256-bit truth (128 pts): max 8.2e−9 **NinjaRS per amplitude** (`ninjars`) 2/128 failed closed, status Unstable, status MasterFailed NinjaRS 69 % **accuracy: 2 pts fail (MasterFailed, Unstable); the call is timed whole**
$gg \to agg$ 5 legs · QCD Rc 1.3e−11 · ML 2.6e−11 · OL 4.4e−11 · partial · 3 pts	**32.0 s** ×38 cold 44.4 s · 2.08 GiB peak warm 2.08 GiB peak · plan 0.0460 MB Rc 836 ms	**23.1 ms** ×0.33 **OL 70.2 ms** · Rc 91.5 ms · ML 149 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.25 on TIRust **TIRust** (`tirust`) numerator 54 % vs 256-bit truth (128 pts): max 3.3e−6 **NinjaRS per amplitude** (`ninjars`) 98.3 ms ×1.4 NinjaRS 78 % **accuracy: 1/128 pts above 1.0e−5, worst 2.0e−4, also in the checked mode**
$gg \to zgg$ 5 legs · QCD Rc 2.9e−6 · ML 2.9e−6 · partial · 3 pts	**23.6 s** ×14 cold 30.1 s · 2.18 GiB peak warm 2.17 GiB peak · plan 0.0520 MB Rc 1.72 s	**72.7 ms** ×0.57 **Rc 127 ms** · ML 251 ms · OL unavailable: no OpenLoops loop-squared library serves g g > z g g (none in the public process list of OpenLoops 2.1.5) $B = 128$ · 3 runtimes ×1.1 pass ×0.44 on TIRust **TIRust** (`tirust`) numerator 41 % **accuracy: 3/128 pts above 1.0e−5, worst 3.1e−5, all served by TIRust's Passarino-Veltman last resort** **NinjaRS per amplitude** (`ninjars`) 330 ms ×2.6 NinjaRS 78 % vs 256-bit truth (128 pts): max 6.2e−6
$u\bar{u} \to hgg$ 5 legs · QCD Rc 7.7e−10 · ML 7.7e−10 · OL 7.8e−10 · partial · 3 pts	**12.1 s** ×13 cold 17.3 s · 0.65 GiB peak warm 0.63 GiB peak · plan 0.0110 MB Rc 960 ms	**2.92 ms** ×1.9 **OL 1.58 ms** · Rc 1.95 ms · ML 3.08 ms $B = 128$ · 3 runtimes ×1.5 <5 ms: not gated on TIRust **TIRust** (`tirust`) TIR moments 40 % vs 256-bit truth (128 pts): max 3.7e−9 **NinjaRS per amplitude** (`ninjars`) 2.39 ms ×1.5 NinjaRS 47 % vs 256-bit truth (128 pts): max 3.5e−8
$u\bar{u} \to d\bar{d}h$ 5 legs · QCD OL 8.1e−11 · Rc 1.2e−10 · ML 1.2e−10 · pass · 3 pts	**12.1 s** ×41 cold 17.5 s · 0.65 GiB peak warm 0.63 GiB peak · plan 0.0100 MB Rc 293 ms	**2.46 ms** ×26 **Rc 0.0944 ms** · ML 0.112 ms · OL 0.205 ms $B = 128$ · 3 runtimes ×3.1 <5 ms: not gated on TIRust **TIRust** (`tirust`) glue 91 % vs 256-bit truth (128 pts): max 4.1e−10 **NinjaRS per amplitude** (`ninjars`) 0.991 ms ×11 glue 89 % vs 256-bit truth (128 pts): max 4.3e−9

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
legs · corrections; largest deviation per reference (AN: analytic result) · verdict · points	public `Generator`, warm cache (cold: empty cache); peak memory; provider generation	median per point at $B = 128$ over fresh runtimes (× spread); paired provider times; gate chip on TIRust; backend (`reduction=` value); stage bars: inputs, numerator, scatter, NinjaRS, TIR moments, contraction, masters, glue (separate instrumented run, not the timed call)
$2 \to 4$		
$gg \to hhgg$ 6 legs · QCD Rc 2.0e−9 · ML 2.8e−9 · OL 1.2e−7 · partial · 4 pts	**55.0 s** ×140 cold 64.3 s · 2.26 GiB peak warm 2.23 GiB peak · plan 0.0470 MB Rc 393 ms	**331 ms** ×1.7 **Rc 196 ms** · OL 346 ms · ML 408 ms · $B = 128$ · 3 runtimes ×1.1 open ×1.3 on TIRust **TIRust** (`tirust`) TIR moments 78 % vs 256-bit truth (32 pts): max 7.1e−7 **NinjaRS per amplitude** (`ninjars`) 600 ms ×3.1 NinjaRS 87 % vs 256-bit truth (32 pts): max 7.0e−7
$gg \to hggg$ 6 legs · QCD Rc 2.2e−11 · ML 3.0e−11 · OL 4.2e−11 · partial · 3 pts	**111 s** ×32 cold 111 s · 2.93 GiB peak warm 2.93 GiB peak · plan 0.100 MB Rc 3.46 s	**257 ms** ×0.35 **Rc 730 ms** · OL 759 ms · ML 1.41 s · $B = 128$ · 3 runtimes ×1.2 pass ×0.27 on TIRust **TIRust** (`tirust`) numerator 43 % vs 256-bit truth (32 pts): max 5.9e−8 **NinjaRS per amplitude** (`ninjars`) 950 ms ×1.3 NinjaRS 75 % **accuracy: 2/32 pts above 1.0e−5, worst 4.2e−5, also in the checked mode**
$gg \to e^+e^-\mu^+\mu^-$ 6 legs · QCD OL 2.7e−13 · ML 3.4e−13 · pass · 3 pts	**26.4 s** ×97 cold 40.4 s · 0.78 GiB peak warm 0.78 GiB peak · plan 0.0380 MB Rc 271 ms	**3.57 ms** ×0.37 Rc 3.61 ms · **OL 9.66 ms** · ML 21.3 ms · $B = 128$ · 3 runtimes ×1.0 Rc not matched (values excluded): ×0.99 to it <5 ms: not gated on TIRust **TIRust** (`tirust`) glue 38 % vs 256-bit truth (32 pts): max 5.4e−12 **NinjaRS per amplitude** (`ninjars`) 5.58 ms ×0.58 NinjaRS 40 % vs 256-bit truth (32 pts): max 6.5e−10
$gg \to aagg$ 6 legs · QCD Rc 4.8e−10 · OL 1.4e−9 · partial · 3 pts	**233 s** ×26 cold 220 s · 3.81 GiB peak warm 3.80 GiB peak · plan 0.112 MB Rc 8.83 s	**720 ms** ×0.31 **OL 2.30 s** · Rc 2.93 s · ML 4.81 s · $B = 128$ · 3 runtimes ×1.2 pass ×0.24 on TIRust **TIRust** (`tirust`) numerator 45 % **TIRust vs NinjaRS (128 pts; no 256-bit truth within the row's time budget): max 7.0e−2, 23 pts above 1.0e−5; vs 256-bit truth at its 2 largest-difference pts: TIRust 1.3e−4, NinjaRS 7.6e−2** **NinjaRS per amplitude** (`ninjars`) 5.50 s ×2.4 no instrumented run
$gg \to zzzz$ 6 legs · QCD Rc 5.2e−6 · partial · 3 pts	**236 s** ×8.7 cold 238 s · 3.83 GiB peak warm 3.78 GiB peak · plan 0.0630 MB Rc 27.0 s	***failed closed*** `tirust` (TIRust): 4/128 failed closed, TIRust status Unstable **Rc 6.19 s** · ML 9.84 s · OL unavailable: no OpenLoops loop-squared library serves g g > z z z z (none in the public process list of OpenLoops 2.1.5) no verdict: TIRust failed closed 4/128 on TIRust · NinjaRS per amplitude: 16/128 failed closed, status Unstable
$2 \to 5$		
$gg \to e^+e^-\mu^+\mu^-g$ or $gg \to hgggg$ 7 legs · QCD OL 5.2e−12 · ML 6.6e−12 · pass · 3 pts	**112 s** ×89 cold 110 s · 2.88 GiB peak warm 2.88 GiB peak · plan 0.0530 MB Rc 1.26 s	**49.8 ms** ×0.14 Rc 116 ms · **OL 344 ms** · ML 777 ms · $B = 128$ · 3 runtimes ×1.1 Rc not matched (values excluded): ×0.43 to it pass ×0.11 on TIRust **TIRust** (`tirust`) numerator 44 % vs 256-bit truth (16 pts): max 8.3e−9 **NinjaRS per amplitude** (`ninjars`) 209 ms ×0.61 NinjaRS 78 % vs 256-bit truth (16 pts): max 7.0e−9
EFT		
EFT process at loop tensor rank > N+2 (HC_NLO_X0 or SMEFT) EFT · TIRust only · execution check	***not timed*** execution check only: no reference, benchmark or timing, since at tensor rank > N+2 the R2 terms would need a dimension-8 model, which does not exist; evaluated through TIRust with two dimension-6 insertions no reference provider	



Not drawn: 2 of the 42 rows of Table 2 (2 no pyAmpliCol time at B = 128). Drawn with an unmatched provider (not a reference, sets no limit): $gg \to e^+e^-\mu^+\mu^-$ (Recola), $gg \to e^+e^-\mu^+\mu^-g$ (Recola).

Figure 1: **Batched time per point against the gate** (finite part, $B = 128$, log scale). Above the rule: the full-colour times of the rows of Table 1 with a matched provider. Below it: the rows of Table 2 with a pyAmpliCol time and a matched provider time.

Table 3: **TIRust and NinjaRS**, the two reduction backends, both with the unchecked setting of Table 2: time per point at $B = 128$ with a stage bar from a separate instrumented run; N ÷ T = NinjaRS time ÷ TIRust time. **Loop-induced:** the timed rows of Table 2 (not shown: 1 row without an unchecked time). "vs truth": the largest relative error against TIRust at 256 bits, in red above 10^{-5}; without a truth, the TIRust–NinjaRS difference, once over both columns. Agreement check: the binary64 TIRust–NinjaRS check of the checked default on the validation points (verdict; T k/n: points TIRust served; the largest finite-part difference of the two backends). Two known TIRust limitations of this release make verdicts partial: a binary64 point whose last-resort reduction fails its accuracy check is refused (a default call takes it from NinjaRS, an explicit `reduction="tirust"` returns it as best-effort, e.g. on $gg \to aag$), and in double-double TIRust has no alternative method for an ill-conditioned integral, so some points have no double-double TIRust value. **Born-interfered:** the $2 \to 3$ and $2 \to 4$ rows of Table 1, with the largest TIRust–NinjaRS difference at the timed points (no truth; in red above 10^{-5}); no instrumented run, so no stage bar.

Process	TIRust	NinjaRS	N ÷ T	T vs truth	N vs truth	Agreement check
Loop-induced ($\lvert A_{\text{loop}}\rvert^2$) · stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue						
$gg \to h$	**0.345 ms** glue 85 %	**0.363 ms** glue 80 %	×1.1	1.2e−13	1.2e−12	pass (T 3/3) 3.2e−13; pass (T 3/3) 3.5e−13
$h \to gg$	**0.368 ms** glue 84 %	**0.386 ms** glue 80 %	×1.0	1.0e−13	1.1e−12	pass (T 3/3) 1.6e−13
$h \to aa$	**0.668 ms** inputs 70 %	**0.717 ms** inputs 74 %	×1.1	2.4e−14	4.2e−13	pass (T 3/3) 5.6e−14
$h \to za$	**1.000 ms** inputs 68 %	**1.06 ms** inputs 72 %	×1.1	1.2e−14	1.6e−12	pass (T 3/3) 2.8e−14
$aa \to h$	**0.708 ms** inputs 73 %	**0.742 ms** inputs 74 %	×1.0	1.4e−14	1.6e−13	pass (T 3/3) 5.5e−14
$gg \to aa$	**1.13 ms** numerator 47 %	**3.54 ms** NinjaRS 54 %	×3.1	6.6e−13	2.4e−13	pass (T 3/3) 1.6e−14; pass (T 3/3) 1.5e−14
$gg \to zz$	**2.36 ms** numerator 52 %	**6.10 ms** NinjaRS 61 %	×2.6	5.5e−12	8.8e−11	pass (T 3/3) 4.2e−12
$gg \to w^+w^-$	**1.82 ms** numerator 46 %	**8.06 ms** NinjaRS 66 %	×4.4	1.2e−12	5.2e−12	pass (T 3/3) 1.7e−13
$gg \to za$	**1.34 ms** numerator 49 %	**3.82 ms** NinjaRS 60 %	×2.9	6.8e−13	6.4e−12	pass (T 3/3) 1.6e−14
$gg \to hh$	**0.387 ms** glue 48 %	**0.564 ms** NinjaRS 41 %	×1.5	3.4e−9	8.8e−11	pass (T 3/3) 3.6e−14
$gg \to hz$	**0.729 ms** glue 37 %	**1.26 ms** NinjaRS 35 %	×1.7	1.4e−12	1.4e−12	pass (T 3/3) 8.3e−14
$gg \to hg$	**1.02 ms** glue 44 %	**1.66 ms** NinjaRS 44 %	×1.6	2.4e−14	7.3e−13	pass (T 3/3) 5.4e−16
$u\bar{u} \to hg$	**0.487 ms** glue 86 %	**0.532 ms** glue 81 %	×1.1	1.1e−14	1.5e−13	pass (T 3/3) 1.8e−15
$ug \to hu$	**0.297 ms** glue 84 %	**0.315 ms** glue 84 %	×1.1	1.2e−14	1.1e−13	pass (T 3/3) 3.0e−15
$gg \to zg$	**3.40 ms** glue 42 %	**12.9 ms** NinjaRS 66 %	×3.8	9.4e−13	9.7e−12	pass (T 3/3) 9.2e−14
$gg \to ag$	**1.15 ms** numerator 47 %	**3.09 ms** NinjaRS 65 %	×2.7	2.3e−13	1.7e−12	pass (T 3/3) 1.1e−13
$e^+e^- \to ha$	**1.86 ms** inputs 73 %	**1.98 ms** inputs 75 %	×1.1	3.8e−14	1.9e−13	pass (T 3/3) 9.5e−15
$aa \to aa$	**1.99 ms** inputs 63 %	**6.78 ms** inputs 62 %	×3.4	1.8e−14	4.9e−14	pass (T 3/3) 6.6e−15
$aa \to zz$	**4.28 ms** inputs 58 %	**11.8 ms** inputs 57 %	×2.8	1.5e−13	7.4e−13	pass (T 3/3) 4.9e−15
$aa \to hh$	**1.10 ms** inputs 74 %	**1.55 ms** inputs 74 %	×1.4	8.8e−11	9.1e−11	pass (T 3/3) 2.1e−14
$e^+e^- \to hh$	**1.00 ms** inputs 81 %	**1.02 ms** inputs 80 %	×1.0	1.3e−10	1.3e−10	pass (T 3/3) 1.2e−14
$h \to ggg$	**1.10 ms** glue 40 %	**1.33 ms** NinjaRS 44 %	×1.2	6.7e−13	1.5e−12	pass (T 3/3) 1.9e−13
$z \to ggg$	**4.17 ms** numerator 47 %	**11.1 ms** NinjaRS 64 %	×2.7	5.5e−11	3.8e−10	pass (T 3/3) 7.4e−12
$e^+e^- \to ggg$	**5.21 ms** glue 41 %	**9.98 ms** NinjaRS 59 %	×1.9	3.5e−7	3.2e−10	pass (T 3/3) 3.5e−7
$gg \to hhh$	**4.60 ms** TIR moments 65 %	**5.36 ms** NinjaRS 57 %	×1.2	7.6e−7	**4.5e64 (2/128 > 1.0e−5)**	pass (T 3/3) 6.6e−13
$gg \to hgg$	**11.2 ms** numerator 43 %	**22.5 ms** NinjaRS 63 %	×2.0	9.2e−10	8.0e−10	pass (T 3/3) 2.7e−10
$gg \to zzg$	**76.0 ms** numerator 45 %	**460 ms** NinjaRS 82 %	×6.1	**1.2e−2 (18/128 > 1.0e−5)**	5.0e−8	pass (T 3/3) 1.8e−8
$gg \to aag$	**24.4 ms** numerator 58 %	**170 ms** NinjaRS 79 %	×7.0	7.0e−6	**1.0e−3 (1/128 > 1.0e−5)**	partial (T 1/3) 2.5e−7
$gg \to w^+w^-g$	**65.0 ms** TIR moments 43 %	**320 ms** NinjaRS 81 %	×4.9	**1.4e−2 (24/128 > 1.0e−5)**	6.2e−8	partial (T 2/3) 8.2e−9
$gg \to zzz$	**43.3 ms** numerator 68 %	failed closed: 20/128 failed closed, status Unstable	—	2.3e−8	—	pass (T 3/3) 5.3e−11

Process	TIRust	NinjaRS	N ÷ T	T vs truth	N vs truth	Agreement check
$gg \to hzg$	**14.2 ms** numerator 54 %	failed closed: 2/128 failed closed, status Unstable, status MasterFailed	—	8.2e−9	—	pass (T 3/3) 3.8e−13
$gg \to agg$	**23.1 ms** numerator 54 %	**98.3 ms** NinjaRS 78 %	×4.3	3.3e−6	**2.0e−4 (1/128 > 1.0e−5)**	partial (T 1/3) 4.5e−8
$gg \to zgg$	**72.7 ms** numerator 41 %	**330 ms** NinjaRS 78 %	×4.5	**3.1e−5 (3/128 > 1.0e−5)**	6.2e−6	pass (T 3/3) 2.9e−6
$u\bar{u} \to hgg$	**2.92 ms** TIR moments 40 %	**2.39 ms** NinjaRS 47 %	×0.82	3.7e−9	3.5e−8	pass (T 3/3) 3.1e−11
$u\bar{u} \to d\bar{d}h$	**2.46 ms** glue 91 %	**0.991 ms** glue 89 %	×0.40	4.1e−10	4.3e−9	pass (T 3/3) 1.3e−10
$gg \to hhgg$	**331 ms** TIR moments 78 %	**600 ms** NinjaRS 87 %	×1.8	7.1e−7	7.0e−7	partial (T 2/3) 3.3e−9; pass (T 1/1) 4.9e−10
$gg \to hggg$	**257 ms** numerator 43 %	**950 ms** NinjaRS 75 %	×3.7	5.9e−8	**4.2e−5 (2/32 > 1.0e−5)**	partial (T 2/3) 3.3e−11
$gg \to e^+e^-\mu^+\mu^-$	**3.57 ms** glue 38 %	**5.58 ms** NinjaRS 40 %	×1.6	5.4e−12	6.5e−10	pass (T 3/3) 2.6e−13
$gg \to aagg$	**720 ms** numerator 45 %	**5.50 s**	×7.6	**TIRust vs NinjaRS (no truth): 7.0e−2 (23/128 > 1.0e−5); vs truth at 2 pts: T 1.3e−4, N 7.6e−2**		partial (T 0/3)
$gg \to zzzz$	failed closed: 4/128 failed closed, TIRust status Unstable	failed closed: 16/128 failed closed, status Unstable	—	—	—	partial (T 2/3) 8.1e−7
$gg \to e^+e^-\mu^+\mu^-g$	**49.8 ms** numerator 44 %	**209 ms** NinjaRS 78 %	×4.2	8.3e−9	7.0e−9	pass (T 3/3) 6.9e−12
Born-interfered (Table 1) · agreement check over the validation set: partial (85 pass, 7 not run, 115 partial, 9 pass (lax))						
$u\bar{u} \to d\bar{d}g$	**2.41 ms**	**2.29 ms**	×0.95		largest |T − N| ÷ max(|T|, |N|) 1.4e−9 over 128 points	
$u\bar{u} \to u\bar{u}g$	**2.56 ms**	**3.45 ms**	×1.3		largest |T − N| ÷ max(|T|, |N|) 4.7e−6 over 128 points	
$u\bar{u} \to e^-e^+g$	**0.639 ms**	**0.610 ms**	×0.95		largest |T − N| ÷ max(|T|, |N|) 8.2e−12 over 128 points	
$u\bar{u} \to ggg$	**9.92 ms**	**6.47 ms**	×0.65		largest |T − N| ÷ max(|T|, |N|) 1.8e−9 over 128 points	
$gg \to ggg$	**32.6 ms**	**25.9 ms**	×0.79		**largest |T − N| ÷ max(|T|, |N|) 2.1e−3 over 128 points**	
$u\bar{u} \to e^-e^+gg$	**11.3 ms**	**4.94 ms**	×0.44		largest |T − N| ÷ max(|T|, |N|) 2.1e−8 over 128 points	
$gg \to t\bar{t}gg$	**410 ms**	**129 ms**	×0.31		largest |T − N| ÷ max(|T|, |N|) 2.7e−8 over 128 points	
$u\bar{u} \to gggg$	**286 ms**	**81.7 ms**	×0.29		largest |T − N| ÷ max(|T|, |N|) 5.6e−6 over 128 points	
$gg \to gggg$	**1.33 s**	**621 ms**	×0.47		**largest |T − N| ÷ max(|T|, |N|) 1.8e−3 over 128 points**	

Missing or incomplete

(generated from the tables' data; T*n* = Table *n*; gate lines: time ÷ limit and pyAmpliCol's largest stage; partial: a check completed on only part of the validation points)

T1 · no value: leading colour: not supported (refused by design: no leading-colour expansion for electroweak corrections) — $gg \to t\bar{t}e^-e^+g$
T1 · no value: no matched provider time (no gate) — $gg \to ggggg$
T2 · no value: time: failed closed (4/128 failed closed, TIRust status Unstable) — $gg \to zzzz$
T2 · no value: NinjaRS per amplitude time: failed closed (16/128 failed closed, status Unstable) — $gg \to zzzz$
T2 · no value: not timed: execution check only: no reference, benchmark or timing, since at tensor rank > N+2 the R2 terms would need a dimension-8 model, which does not exist — EFT process at loop tensor rank > N+2 (HC_NLO_X0 or SMEFT)
T2 · gate: open on TIRust — $e^+e^- \to ggg$ (×0.90, glue 41 %), $gg \to hhgg$ (×1.3, TIR moments 78 %)
T2 · validation: partial: rotation (checked default) — $h \to ggg$, $e^+e^- \to ggg$
T2 · validation: partial: pole cancellation, Ward identity, rotation (checked default) — $z \to ggg$, $u\bar{u} \to hgg$
T2 · validation: partial: pole cancellation, Ward identity, rotation, TIRust-NinjaRS agreement (checked default) — $gg \to hgg$, $gg \to zzg$, $gg \to zgg$
T2 · validation: partial: pole cancellation, Ward identity, rotation, TIRust-NinjaRS agreement; TIRust fails 2 of 3 binary64 validation points (checked default) — $gg \to aag$, $gg \to agg$
T2 · validation: partial: pole cancellation, Ward identity, rotation, TIRust-NinjaRS agreement; TIRust fails 1 of 3 binary64 validation points (checked default) — $gg \to w^+w^-g$, $gg \to hhgg$, $gg \to hggg$, $gg \to zzzz$
T2 · validation: partial: pole cancellation, Ward identity, rotation, TIRust-NinjaRS agreement; TIRust fails 3 of 3 binary64 validation points (checked default) — $gg \to aagg$
T2 · accuracy: NinjaRS per amplitude unchecked: 2/128 points above 1e-05, worst 4.5e+64 — $gg \to hhh$
T2 · accuracy: TIRust unchecked: 18/128 points above 1e-05, worst 1.2e-02 — $gg \to zzg$
T2 · accuracy: NinjaRS per amplitude unchecked: 1/128 points above 1e-05, worst 1.0e-03 — $gg \to aag$
T2 · accuracy: TIRust unchecked: 24/128 points above 1e-05, worst 1.4e-02 — $gg \to w^+w^-g$
T2 · accuracy: NinjaRS per amplitude unchecked: 20 points fail (Unstable) — $gg \to zzz$
T2 · accuracy: NinjaRS per amplitude unchecked: 2 points fail (MasterFailed, Unstable) — $gg \to hzg$
T2 · accuracy: NinjaRS per amplitude unchecked: 1/128 points above 1e-05, worst 2.0e-04 — $gg \to agg$
T2 · accuracy: TIRust unchecked: 3/128 points above 1e-05, worst 3.1e-05 — $gg \to zgg$
T2 · accuracy: NinjaRS per amplitude unchecked: 2/32 points above 1e-05, worst 4.2e-05 — $gg \to hggg$
T2 · accuracy: TIRust vs NinjaRS unchecked, no 256-bit truth (a difference, not an error): 23/128 points differ above 1e-05, largest 7.0e-02; at its 2 largest-difference points against the 256-bit truth: TIRust 1.3e-04, NinjaRS 7.6e-02 — $gg \to aagg$
T3 · accuracy: Born-interfered, TIRust vs NinjaRS unchecked (a difference, not an error): largest 2.1e-03 over 128 points, above 1e-05 — $gg \to ggg$
T3 · accuracy: Born-interfered, TIRust vs NinjaRS unchecked (a difference, not an error): largest 1.8e-03 over 128 points, above 1e-05 — $gg \to gggg$

Table 4: **Gate and stage split at $B = 128$**, ms per point, for every timed row of Table 1: pyAmpliCol's full-colour median, the binding limit and its provider, the band of time ÷ limit (standard error of pyAmpliCol's median plus the provider's run-to-run half-range), the gate chip and pyAmpliCol's time per stage.

Process	ms/pt	Limit	Band	Gate	inputs	numerator	scatter	NinjaRS	masters	glue
$gg \to gg$	**1.44**	—	—	no gate (2→2)	0.241	0.573	0.0871	0.162	0.0527	0.229
$u\bar{u} \to d\bar{d}g$	**3.02**	4.69 OL	0.590–0.698	<5 ms: not gated ×0.64	0.171	1.01	0.657	0.327	0.218	0.614
$u\bar{u} \to u\bar{u}g$	**2.43**	4.96 OL	0.468–0.513	<5 ms: not gated ×0.49	0.158	0.922	0.458	0.290	0.204	0.402
$u\bar{u} \to e^-e^+g$	**0.672**	0.770 Rc	0.858–0.887	<5 ms: not gated ×0.87	0.0492	0.227	0.131	0.0344	0.0236	0.193
$u\bar{u} \to ggg$	**6.73**	29.1 OL	0.202–0.260	pass ×0.23	0.997	2.84	0.668	0.931	0.354	0.871
$gg \to ggg$	**24.6**	222 OL	0.0979–0.123	pass ×0.11	7.53	10.0	0.774	4.49	0.784	1.44
$u\bar{u} \to e^-e^+gg$	**6.14**	24.5 Rc	0.232–0.268	pass ×0.25	0.256	2.57	0.804	0.690	0.330	0.679
$gg \to t\bar{t}gg$	**76.1**	652 OL	0.116–0.118	pass ×0.12	12.8	27.7	1.01	21.6	7.01	5.52
$u\bar{u} \to gggg$	**58.7**	517 OL	0.109–0.118	pass ×0.11	11.4	27.9	1.85	10.2	3.17	3.17
$gg \to gggg$	**463**	6 240 OL	0.0653–0.0832	pass ×0.074	188	165	3.29	72.0	8.68	7.47
$u\bar{u} \to e^-e^+d\bar{d}g$	**14.3**	49.7 Rc	0.274–0.300	pass ×0.29	0.840	6.35	2.00	2.43	1.60	1.39
$u\bar{u} \to e^-e^+ggg$	**43.2**	372 Rc	0.115–0.117	pass ×0.12	1.91	24.9	2.23	8.97	2.89	2.36
$gg \to t\bar{t}e^-e^+g$	**75.4**	944 OL	0.0653–0.0945	pass ×0.080	5.81	33.4	1.18	20.4	6.20	3.06
$gg \to t\bar{t}e^-e^+g$ (EW)	**181**	1 980 OL	0.0854–0.0973	pass ×0.091	12.1	66.9	2.69	54.8	22.6	10.8
$u\bar{u} \to ggggg$	**1 810**	569 000 ML	0.00219–0.00415	pass ×0.0032	367	913	12.3	364	63.7	36.7
$gg \to ggggg$	**5 110**	—	—	no gate (no matched provider)	2 340	2 250	10.3	444	44.7	30.5

Reading the tables

Timing settings. Every time is a median per phase-space point over batches of $B = 128$ distinct points: binary64, finite part, one thread pinned to one core (`taskset`, `RAYON_NUM_THREADS=1`), SymJIT `-O2` evaluators as shipped, no stability check, no rescue. Every setting is a public option; nothing is modified, estimated or extrapolated. The stored $2 \to 6$ rows of Table 1 are outside this protocol (labelled in their rows).

- **Table 1:** `python -m tools.loop_benchmark pyamplicol <row> --inputs native --batches 128`, the public evaluation (values bitwise those of `Runtime.evaluate`) with the default reduction (`reduction_unchecked = false`: NinjaRS runs its per-point acceptance test). Every timed point was accepted (a refused point stops the run), so the default failure policy changes neither values nor times. A paired same-core A/B against `reduction_unchecked = True` on the nine $2 \to 3$ and $2 \to 4$ rows (3 fresh instances per arm, 128 points) gives bitwise-identical values and unchecked ÷ default ×0.92–×1.25 (×1.25: one noisy g g ⟶ g g g sample; the others ×0.92–×1.05).
- **Table 2 and Table 3:** `Runtime.load(artifact, reduction_unchecked=True)`, then `evaluate(points, mu_r2=…, poles="none", on_failure="return", reduction="tirust" | "ninjars")`, with no `stability_checks`, no rescue policy and `kinematics_improvement_binary64` off.
- **Another process:** `pyamplicol loop_generate.toml generate --process="…"`, then `pyamplicol loop_generate.toml profile --runtime --reduction-unchecked --batch-size 128` (master card `examples/loop_generate.toml`) times the same public evaluation on both backends, on flat points instead of the frozen streams and without providers.

Unchecked setting (`reduction_unchecked`, public, off by default). Each backend skips its own acceptance and refusal checks. TIRust keeps its a priori conditioning gates and its method choice, and serves what the method choice refuses by plain Passarino–Veltman (its last resort). NinjaRS runs with zero stability thresholds. pyAmpliCol's own checks are off. A point that no scheme evaluates finitely still fails: an exactly singular pivot, a non-finite output, a master-integral failure, or a NinjaRS point whose tested quantity is exactly zero or NaN (`Unstable`). It fails under every failure policy, since an unchecked runtime has no check left to skip. A call with failed points is timed whole, so it has no time per served point (marker **failed closed**). The setting has no accuracy guarantee; each cell prints its measured accuracy.

Providers. The providers run in their most optimistic settings, on pyAmpliCol's core. OpenLoops 2.1.5 (OL): prebuilt libraries, so no generation time; `expert_mode 1`, `stability_mode 11` (stability system off), `truepoles 0`, and `ir_on 0` on the Born-interfered rows. MadLoop 3.7.0 (ML): `CTModeRun 1` in `MadLoopParams.dat` (no stability test). Recola 2.2.5 with COLLIER (Rc), release build: `set_dynamic_settings_rcl(0)`, `set_compute_ir_poles_rcl(0)`; α_s and the scale are set before its generation, so it times the stream points at the first point's μ_R (6 points per row, against up to 16 for the others). Each optimistic value equals the provider's production value at the timed points. [P]: the provider's production record, kept where the provider's generation for an optimistic run takes longer than 45 minutes.

Table 1. One row per process, grouped by multiplicity. Bold: pyAmpliCol's value. ×: pyAmpliCol ÷ the fastest provider (bold among the provider times). Only run times are coloured: green faster, white at parity, red slower ($\log_2$ scale, clamped at ×1/16 and ×16). Generation is generate + plan export (model loading excluded), with peak memory and plan size; its multiplier is not coloured. LC: leading colour, with no provider ratio; FC: full colour. Grey cells give the reason a value is missing. The ≤ bound of the caption is Recola's run that also evaluates the IR poles. The table's first row names the engine, native build and digest, date and host; a cell measured elsewhere names its origin.

Gate chips (Table 1, Figure 1, Table 4) give time ÷ limit. The limit is the smallest of MadLoop's time and 1.30 × each of OpenLoops' and Recola's. **pass** or **fail**: the whole uncertainty band (Table 4) lies below or above 1; else **open**. Below 5 ms per point a process is not gated ("<5 ms: not gated"); its chip keeps the measured ratio. $2 \to 2$ has no gate. Gates are taken at $B = 128$ only.

Stage bars (one colour per stage; the cell names the largest share; Table 4 lists the stages of Table 1 in ms). inputs: trees, Born and counterterm inputs (loop-induced: the trees and counterterms of each amplitude component). numerator: seeds, fused factors and the dressing of the open-loop objects. scatter: handing the numerator closures to the reduction. NinjaRS: its expansions (the reduction without the masters). masters: the master integrals (oneloopmaster). TIR moments: TIRust's tensor moments, its master integrals included unless recorded apart. contraction: their contraction with the numerator. glue: point parameters, input conversion, transport, squaring, assembly and the rest of the call.

Validation (process cell): the largest deviation $|a - b| \,/\, \max(|a|, |b|, |B|\ \alpha_s/2\pi)$ of the binary64 finite part per reference at the timed points, and the median over points and references (NJet DD: NJet 3.1.2 in double-double). †: the largest lies within pyAmpliCol's binary64 stability estimate at that point. ¶: it is the reference's own: pyAmpliCol's double-double value lies as far from the reference, and its binary64 value lies within its estimate of the double-double one. ‡ $gg \to ggggg$: points 4 and 8 (1.5×10^{-7} and 2.6×10^{-8} from NJet DD) exceed the check's tolerance 2×10^{-8} in binary64: **accepted: binary64 within the estimate, DD = NJet** (estimates E 1.5×10^{-6} and 3.4×10^{-7}; pyAmpliCol's double-double values equal the reference's). The check keeps its tolerance and passes in double-double at these points.

Notes under a value qualify it; the value stays as measured (amber: a known limitation; grey: run conditions). $gg \to t\bar{t}e^-e^+g$, full colour ("model: unpatched G_mu UFO"): timed with the benchmark tool's default copy of the loop_qcd_qed_sm_Gmu UFO model, not with the patched model that pyAmpliCol ships and validated (UFOmodeldb 6ca4b120 with fixes to an R2 coupling, the CKM matrix and a b-mass counterterm); the gate is taken on this timing.

The $2 \to 6$ rows have no provider, so no multiplier, gate or colour. Generation: generate + plan export, with store, artifact size and load; "FC" marks the full-colour generation, LC below it. Timing: binary64, finite part, checks off, one pinned thread, at the largest batch measured. The process cell lists the internal checks (ε^{-2}: double pole against its universal value; E: stability estimate, with the points above 10^{-5}; rescue: stream points the default rescue policy sends up from binary64; Ward on the external gluons; binary64 against double-double; Catani at full colour; NGluon at leading colour on the all-gluon row) and, in grey, those not run or not asserted.

Table 2 reads as Table 1, for $|A_{\text{loop}}|^2$ without a Born. One row per process of the loop-induced validation set, by multiplicity; its 11 validation-only rows (backend unit tests, processes that vanish, identity checks, observables and the complex-mass scheme) are not listed. The cell is TIRust with the unchecked setting. Its time is the median over 3 fresh runtimes, created in rotated order, of each runtime's median over rounds; the spread (×, largest ÷ smallest) stands beside B. `auto` with the setting is TIRust, bitwise and without fallback (paired on $gg \to hgg$). Beneath: NinjaRS with the same setting, with its time and multiplier where it served every point, else its failed points and their cause. Stage bars come from a **separate instrumented run** (same setting, points, B and core), never the timed call. Each backend's line ends with its accuracy on the row's first stream points against TIRust at 256 bits: the largest relative error, or in red the points above 10^{-5}, the worst and how they were served, and the failed points. Without a 256-bit truth the TIRust–NinjaRS difference is shown once.

Multiplier and colour use the fastest matched provider. A provider whose values are not a reference (Recola's on-shell widths against the others' fixed widths, on $gg \to e^+e^-\mu^+\mu^-$ and $gg \to e^+e^-\mu^+\mu^-g$) is timed and shown but never matched: it sets no colour and no limit; the cell gives the multiplier to it. Generation (not coloured): the public `Generator` from the process card, warm (a second generation in the same cache: generate, the exact helicity-zero filter and plan export) in the value, cold (an empty cache: the model is also compiled and prepared) beneath, each with its peak memory.

Gate: none for four legs or fewer ($2 \to 1$, $1 \to 2$, $2 \to 2$, $1 \to 3$); from $2 \to 3$ upward the gate of Table 1, on TIRust ("on TIRust"), "<5 ms: not gated" below 5 ms per point, none without a matched provider. Where TIRust failed points, the chip reads "no verdict": there is no time per served point. Grey markers: **not run** (no record, or a run stopped before its timing, with its progress, e.g. a generation beyond 45 min); **failed closed** or **refused** (the requested call failed points or was refused: no time); a provider's reason where it has no matched time.

Data: `data/summary.json` and `data/summary-li.json`, written by `scripts/build_summary.py` from the records in `data/` (`README.md`).